

at the time was that no one was able to isolate what it was that the high frequency traders were doing at the time that was not everyday business. But there was a large market belief that it was the high frequency traders' fault. They served as a nice scapegoat. Especially since it has been reported that over the previous few years they all made boat loads of profits; so they must be bad.

The SEC issued a report in September 2010 discussing the flash crash. While the report put forth many theories and hypotheses for its cause, it did not provide ample statistical evidence supporting the theories behind the cause. Many of these potential reasons stated as potential causes of the crash also happened in the past and without any issues at all, so there must have been, as many have stated, a perfect storm. But what really was that perfect storm?

There was also a lot of discussion surrounding the events and people stating that the markets are fragile and the current structure is flawed. Many have stated that regulators need to make some significant and drastic changes to the structure. Some have stated that spreads need to increase to nickels while others stated that spreads need to decrease to sub-penny.

The important take-away points from the flash crash for algorithmic trading:

- Algorithmic and electronic trading is exponentially quicker than humans transacting orders manually. The nice thing about algorithms is that they will do exactly what you program them to do, and the not-so-nice thing about algorithms is that they will do exactly what you program them to do. So if you do not program every potential event into its logic there will be times when there are unforeseen market events and this could lead to dire results.
- Specialists and market makers always played a very important role in financial markets regarding price discovery. While this process failed during the program crash of 1987, I do not recall any other event since then leading up to the flash crash where there was such a large market decline. Specialists and market makers have been halting orders when appropriate and spending time to understand what is going on. We are now in a system where there is no single specialist and/or market maker controlling stocks and halting trading when necessary.
- Developers need to incorporate safeguards into algorithms to protect investors from the ominous days. However, these safeguards cannot be too strict because they may exclude investors from transacting in a rapidly moving market where there are no issues.